

Question	Answer	Marks
1(a)	$F = (Gm_1m_2) / r^2$	M1
	where G is the gravitational constant	A1
1(b)	gravitational force provides the centripetal force	B1
	$mR\omega^2 = GMm / R^2$ and $\omega = 2\pi / T$ or $mv^2 / R = GMm / R^2$ and $v = 2\pi R / T$ or $4\pi^2mR / T^2 = GMm / R^2$	M1
	correct completion of algebra to get $T^2 = (4\pi^2 / GM) R^3$, with identification of $(4\pi^2 / GM)$ as k	A1
1(c)(i)	$(24 \times 3600)^2 = (4\pi^2 \times R^3) / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$	C1
	$R = 4.2 \times 10^7 \text{ m}$	A1
1(c)(ii)	(orbit) must be above the Equator	B1
	(direction) must be from west to east	B1

9702/43

Cambridge International AS & A Level – Mark Scheme

October/November 2022

PUBLISHED

Question	Answer	Marks
2(c)	The first 5 minutes	20